SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

LIST OF EXPERIMENTS

SUB CORE & NAME:

CSA1703- ARTIFICIAL INTELLIGENCE

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E.Ragul

1. 8 Puzzle Problem

Aim

To solve the 8-Puzzle problem using the A* Search algorithm in Python.

Algorithm

1. Define the initial state and goal state of the puzzle.
2. Calculate the heuristic value (number of misplaced tiles).
3. Generate all possible moves (up, down, left, and right).
4. Select the state with the lowest cost using the A* search algorithm.
5. Repeat the process until the goal state is reached.
6. Display the sequence of moves from the initial state to the goal state.

Program

```
from queue import PriorityQueue

goal = [[1,2,3],
        [4,5,6],
        [7,8,0]]

def heuristic(state):
    count = 0
    for i in range(3):
        for j in range(3):
            if state[i][j] != 0 and state[i][j] != goal[i][j]:
                count += 1
    return count

def find_blank(state):
    for i in range(3):
        for j in range(3):
            if state[i][j] == 0:
                return i, j

def copy(state):
    return [row[:] for row in state]

def neighbors(state):
    x, y = find_blank(state)
    moves = [(1,0),(-1,0),(0,1),(0,-1)]
    result = []

    for dx, dy in moves:
        nx, ny = x+dx, y+dy
        if 0 <= nx < 3 and 0 <= ny < 3:
            new = copy(state)
            new[x][y], new[nx][ny] = new[nx][ny], new[x][y]
```

```
            result.append(new)
    return result

start = [[1,2,3],
         [4,0,6],
         [7,5,8]]

pq = PriorityQueue()
pq.put((heuristic(start), start))
visited = []

while not pq.empty():
    h, state = pq.get()

    if state == goal:
        print("Goal State Reached")
        for row in state:
            print(row)
        break

    if state not in visited:
        visited.append(state)
        for next_state in neighbors(state):
            if next_state not in visited:
                pq.put((heuristic(next_state), next_state))
```

Input

Initial State

1 2 3
4 0 6
7 5 8

Output

Goal State Reached

[1, 2, 3]
[4, 5, 6]
[7, 8, 0]

Result

Thus, the 8-Puzzle problem was solved successfully using the *A Search algorithm* in Python.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

File Edit Format Run Options Window Help

```
from queue import PriorityQueue

goal = [[1,2,3],
        [4,5,6],
        [7,8,0]]

def heuristic(state):
    count = 0
    for i in range(3):
        for j in range(3):
            if state[i][j] != 0 and state[i][j] != goal[i][j]:
                count += 1
    return count

def find_blank(state):
    for i in range(3):
        for j in range(3):
            if state[i][j] == 0:
                return i, j

def copy(state):
    return [row[:] for row in state]

def neighbors(state):
    x, y = find_blank(state)
    moves = [(1,0),(-1,0),(0,1),(0,-1)]
    result = []

    for dx, dy in moves:
        nx, ny = x+dx, y+dy
        if 0 <= nx < 3 and 0 <= ny < 3:
            new = copy(state)
            new[x][y], new[nx][ny] = new[nx][ny], new[x][y]
            result.append(new)
    return result

start = [[1,2,3],
         [4,0,6],
         [7,5,8]]
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
Goal State Reached
[1, 2, 3]
[4, 5, 6]
[7, 8, 0]
>>>
```

2. 8 Queen Problem

Aim

To develop a Python program to solve the **8-Queen Problem** using the **Backtracking Algorithm** by placing eight queens on an 8×8 chessboard such that no two queens attack each other.

Algorithm

1. Create an empty 8×8 chessboard.
2. Start placing queens from the first row.
3. Check whether the current position is safe:
    - No queen in the same column.
    - No queen in the left diagonal.
    - No queen in the right diagonal.
4. If the position is safe, place the queen.
5. Recursively place queens in the next row.
6. If no safe position exists, backtrack and try another column.
7. Repeat until all eight queens are placed.
8. Display the final chessboard arrangement.

Program

```
N = 8

def is_safe(board, row, col):
    for i in range(row):
        if board[i] == col:
            return False
        if abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(board, row):
    if row == N:
        return True

    for col in range(N):
        if is_safe(board, row, col):
            board[row] = col
            if solve(board, row + 1):
                return True
```

```
            board[row] = -1
    return False

board = [-1] * N

if solve(board, 0):
    print("Solution:")
    for i in range(N):
        for j in range(N):
            if board[i] == j:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No Solution Exists")
```

Input

No input is required.

Output

```
Solution:

Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

Result

Thus, the **8-Queen Problem** was solved successfully using the **Backtracking Algorithm** in Python.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

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```
N = 8

def is_safe(board, row, col):
    for i in range(row):
        if board[i] == col:
            return False
        if abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(board, row):
    if row == N:
        return True

    for col in range(N):
        if is_safe(board, row, col):
            board[row] = col
            if solve(board, row + 1):
                return True
            board[row] = -1
    return False

board = [-1] * N

if solve(board, 0):
    print("Solution:")
    for i in range(N):
        for j in range(N):
            if board[i] == j:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No Solution Exists")
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
Solution:
Q . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
>>>
```

3. Water Jug Problem

Aim

To write a Python program to solve the **Water Jug Problem** using the **Breadth-First Search (BFS)** algorithm to obtain the required quantity of water.

Algorithm

1. Start with both jugs empty.
2. Represent each state as (Jug1, Jug2).
3. Use a queue to store the states to be explored.
4. Generate all possible operations:
    - Fill Jug 1.
    - Fill Jug 2.
    - Empty Jug 1.
    - Empty Jug 2.
    - Pour water from Jug 1 to Jug 2.
    - Pour water from Jug 2 to Jug 1.
5. Mark each visited state to avoid repetition.
6. Repeat until the target amount of water is obtained.
7. Display the sequence of states and the final solution.

Program

```
from collections import deque

def water_jug(jug1, jug2, target):
    visited = set()
    queue = deque([(0, 0)])

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
            continue

        visited.add((x, y))
        print((x, y))

        if x == target or y == target:
            print("Target Reached")
            return
```

```
        # Fill either jug
        queue.append((jug1, y))
        queue.append((x, jug2))

        # Empty either jug
        queue.append((0, y))
        queue.append((x, 0))

        # Pour Jug1 -> Jug2
        t = min(x, jug2 - y)
        queue.append((x - t, y + t))

        # Pour Jug2 -> Jug1
        t = min(y, jug1 - x)
        queue.append((x + t, y - t))

water_jug(4, 3, 2)
```

Input

```
Capacity of Jug 1 = 4 litres
Capacity of Jug 2 = 3 litres
Target = 2 litres
```

Output

```
(0, 0)
(4, 0)
(0, 3)
(1, 3)
(3, 0)
(1, 0)
(3, 3)
(0, 1)
(4, 2)
Target Reached
```

Result

Thus, the **Water Jug Problem** was implemented successfully in Python, and the required **2 litres** of water was obtained

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

File Edit Format Run Options Window Help

```
from collections import deque

def water_jug(jug1, jug2, target):
    visited = set()
    queue = deque([(0, 0)])

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
            continue

        visited.add((x, y))
        print((x, y))

        if x == target or y == target:
            print("Target Reached")
            return

        # Fill either jug
        queue.append((jug1, y))
        queue.append((x, jug2))

        # Empty either jug
        queue.append((0, y))
        queue.append((x, 0))

        # Pour Jug1 -> Jug2
        t = min(x, jug2 - y)
        queue.append((x - t, y + t))

        # Pour Jug2 -> Jug1
        t = min(y, jug1 - x)
        queue.append((x + t, y - t))

water_jug(4, 3, 2)
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
(0, 0)
(4, 0)
(0, 3)
(4, 3)
(1, 3)
(3, 0)
(1, 0)
(3, 3)
(0, 1)
(4, 2)
Target Reached
>>>
```

4. Crypt Arithmetic Problem

Aim

To write a Python program to solve the Crypt Arithmetic Problem by assigning unique digits to letters so that the given arithmetic equation is satisfied.

Algorithm

1. Define the words in the crypt arithmetic problem
2. Assign unique digits (0–9) to each letter.
3. Ensure that the leading letters are not assigned zero.
4. Convert the words into numbers.
5. Check whether the arithmetic equation is satisfied.
6. If a valid assignment is found, display the solution.
7. Stop the program.

Program

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')

for p in permutations(range(10), 8):
    d = dict(zip(letters, p))

    if d['S'] == 0 or d['M'] == 0:
        continue

    SEND = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
    MORE = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
    MONEY = 10000*d['M'] + 1000*d['O'] + 100*d['N'] + 10*d['E'] + d['Y']

    if SEND + MORE == MONEY:
        print("Solution:")
        print(d)
        print(SEND, "+", MORE, "=", MONEY)
        break
```

Input

No input is required.

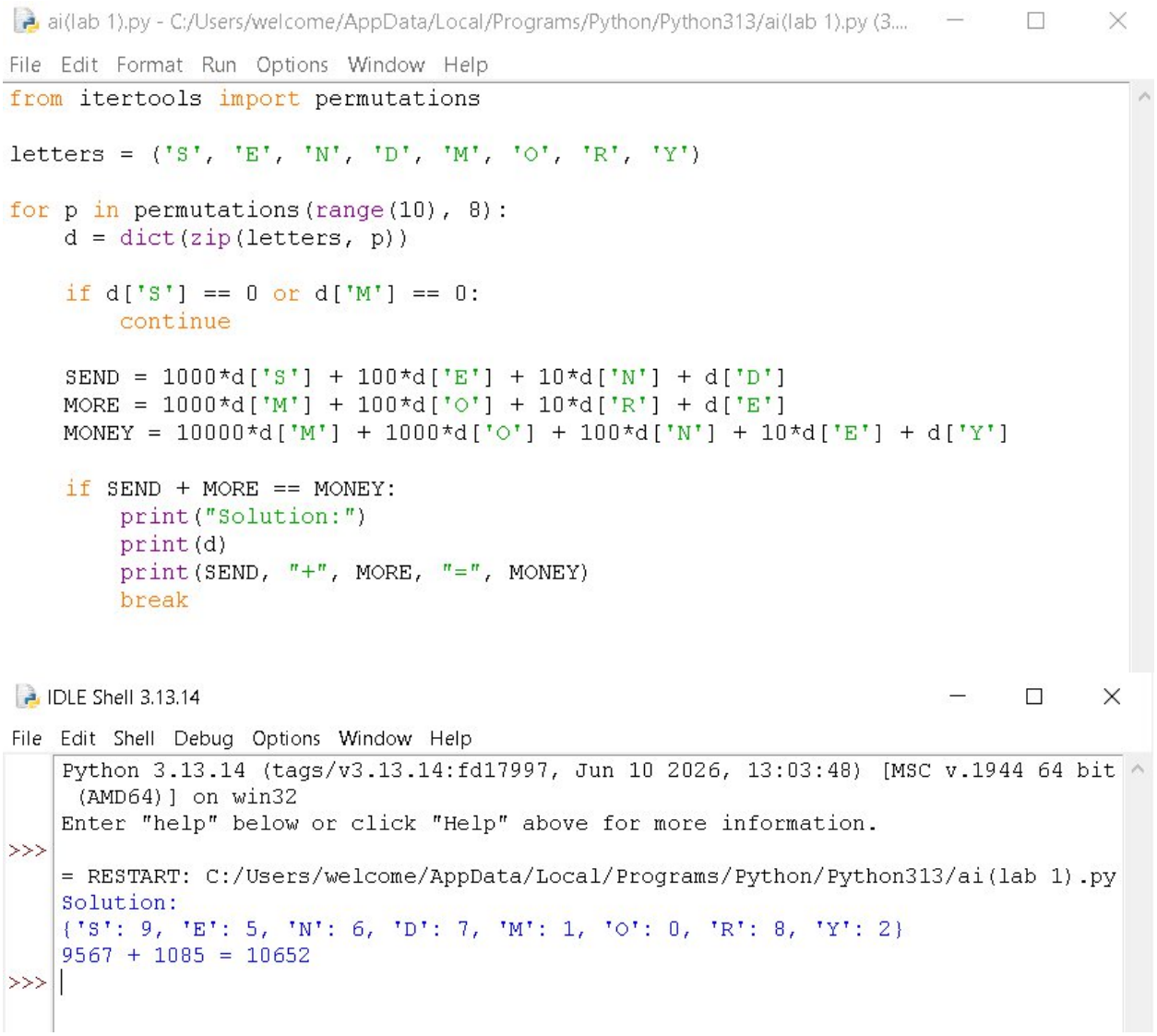
Output

```
Solution:
{'S': 9, 'E': 5, 'N': 6, 'D': 7,
 'M': 1, 'O': 0, 'R': 8, 'Y': 2}

9567 + 1085 = 10652
```

Result

Thus, the **Crypt Arithmetic Problem** was solved successfully using Python by assigning unique digits to letters that satisfy the given arithmetic expression.



5. Missionaries and Cannibals Problem

Aim

To write a Python program to solve the **Missionaries and Cannibals Problem** using **Breadth-First Search (BFS)**.

Algorithm

1. Represent the initial state as **(3 Missionaries, 3 Cannibals, Left Bank)**.
2. Represent the goal state as **(0 Missionaries, 0 Cannibals, Right Bank)**.
3. Define all possible boat moves.
4. Generate valid successor states.
5. Ensure missionaries are never outnumbered by cannibals on either bank.
6. Use BFS to explore all valid states.
7. Stop when the goal state is reached and display the solution.

Program

```
from collections import deque

def is_valid(m, c):
    if m < 0 or c < 0 or m > 3 or c > 3:
        return False
    if (m > 0 and m < c):
        return False
    if (3 - m > 0 and 3 - m < 3 - c):
        return False
    return True

def missionaries_cannibals():
    start = (3, 3, 'L')
    goal = (0, 0, 'R')

    moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]

    queue = deque([start])
    visited = set()

    while queue:
        state = queue.popleft()

        if state in visited:
            continue

        visited.add(state)
        print(state)
        if state == goal:
            print("Goal Reached")
```

```
            return

        m, c, side = state

        for dm, dc in moves:
            if side == 'L':
                nm, nc = m-dm, c-dc
                ns = 'R'
            else:
                nm, nc = m+dm, c+dc
                ns = 'L'

            if is_valid(nm, nc):
                queue.append((nm, nc, ns))

missionaries_cannibals()
```

Input

No input is required.

Output

```
(3, 3, 'L')
(3, 2, 'R')
(3, 1, 'L')
...
(0, 0, 'R')
Goal Reached
```

Result

Thus, the **Missionaries and Cannibals Problem** was solved successfully using the **Breadth-First Search (BFS)** algorithm in Python.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

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```
from collections import deque

def is_valid(m, c):
    if m < 0 or c < 0 or m > 3 or c > 3:
        return False
    if (m > 0 and m < c):
        return False
    if (3 - m > 0 and 3 - m < 3 - c):
        return False
    return True

def missionaries_cannibals():
    start = (3, 3, 'L')
    goal = (0, 0, 'R')

    moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]

    queue = deque([start])
    visited = set()

    while queue:
        state = queue.popleft()

        if state in visited:
            continue

        visited.add(state)
        print(state)

        if state == goal:
            print("Goal Reached")
            return

        m, c, side = state

        for dm, dc in moves:
            if side == 'L':
                nm, nc = m-dm, c-dc
                ns = 'R'
            else:
```

Ln: 47 Col: 24

IDLE Shell 3.13.14

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
(3, 3, 'L')
(3, 2, 'R')
(3, 1, 'R')
(2, 2, 'R')
(3, 2, 'L')
(3, 0, 'R')
(3, 1, 'L')
(1, 1, 'R')
(2, 2, 'L')
(0, 2, 'R')
(0, 3, 'L')
(0, 1, 'R')
(1, 1, 'L')
(0, 2, 'L')
(0, 0, 'R')
Goal Reached
```

6. Vacuum Cleaner Problem

Aim

To write a Python program to simulate a Vacuum Cleaner Agent that cleans all dirty rooms.

Algorithm

1. Initialize the rooms and their status (Dirty/Clean).
2. Start from the first room.
3. Check the status of the current room.
4. If the room is dirty, clean it.
5. Move to the next room.
6. Repeat until all rooms are clean.
7. Display the final status of all rooms.

Program

```
rooms = {'A': 'Dirty', 'B': 'Dirty'}

for room in rooms:
    print("Current Room:", room)

    if rooms[room] == "Dirty":
        print("Cleaning", room)
        rooms[room] = "Clean"

    print(room, ":", rooms[room])

print("\nFinal Room Status:")
for room in rooms:
    print(room, ":", rooms[room])

print("All rooms are clean.")
```

Input

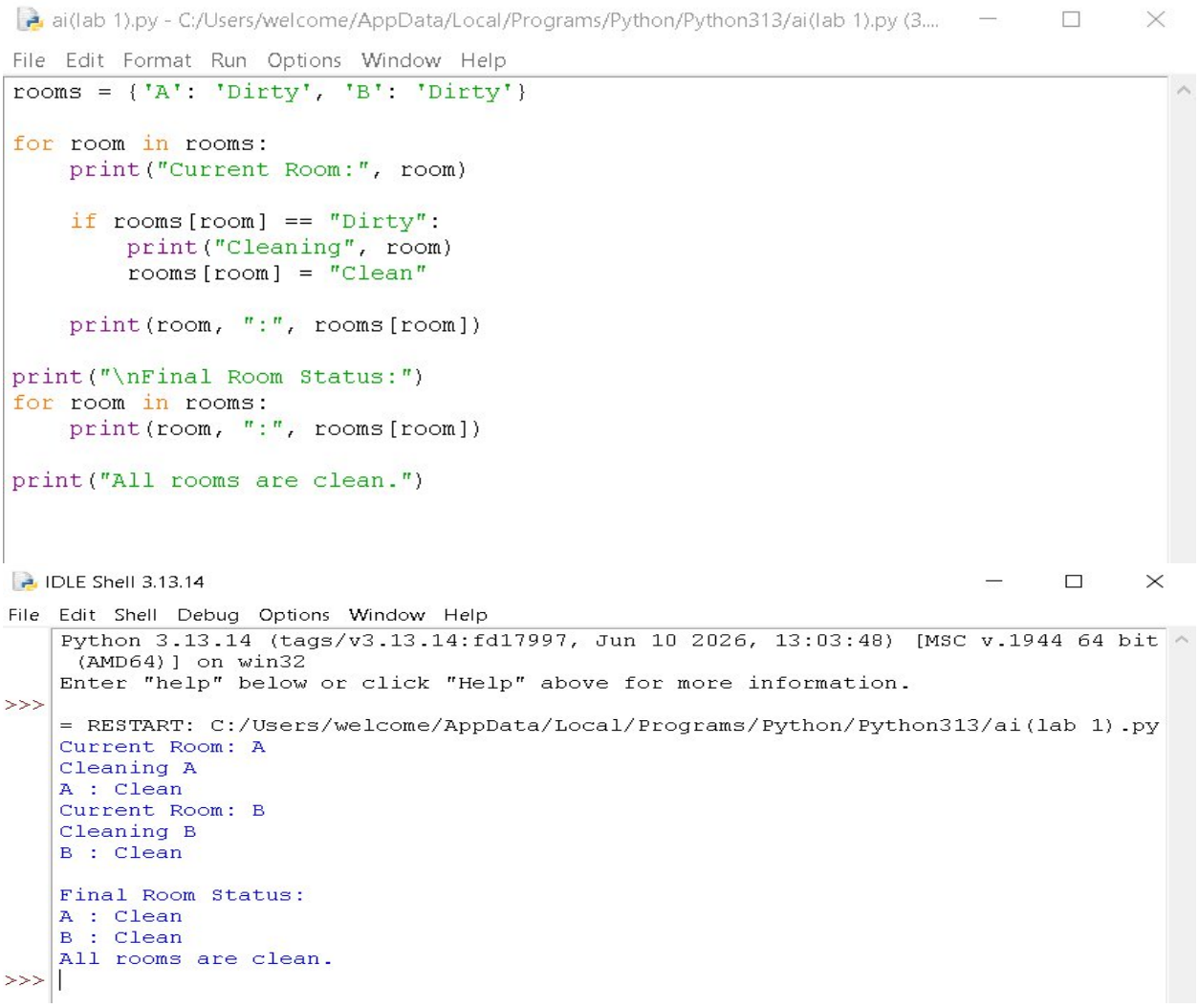
```
Room A = Dirty
Room B = Dirty
```

Output

```
Current Room: A
Cleaning A
A : Clean

Current Room: B
Cleaning B
B : Clean

Final Room Status:
A : Clean
B : Clean

All rooms are clean.
```

Result

Thus, the **Vacuum Cleaner Problem** was implemented successfully in Python, and all dirty rooms were cleaned.


ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....
File Edit Format Run Options Window Help
rooms = {'A': 'Dirty', 'B': 'Dirty'}
for room in rooms:
print("Current Room:", room)
if rooms[room] == "Dirty":
print("Cleaning", room)
rooms[room] = "Clean"
print(room, ":", rooms[room])
print("\nFinal Room Status:")
for room in rooms:
print(room, ":", rooms[room])
print("All rooms are clean.")
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
Current Room: A
Cleaning A
A : Clean
Current Room: B
Cleaning B
B : Clean
Final Room Status:
A : Clean
B : Clean
All rooms are clean.
>>>

7. Breadth First Search (BFS)

Aim

To write a Python program to implement the Breadth First Search (BFS) algorithm for graph traversal.

Algorithm

1. Start from the source node.
2. Mark the source node as visited.
3. Insert the source node into a queue.
4. Remove a node from the queue.
5. Display the node.
6. Visit all unvisited adjacent nodes and insert them into the queue.
7. Repeat until the queue becomes empty.

Program

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def bfs(start):
    queue = deque([start])
    visited.add(start)

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

bfs('A')
```

Input

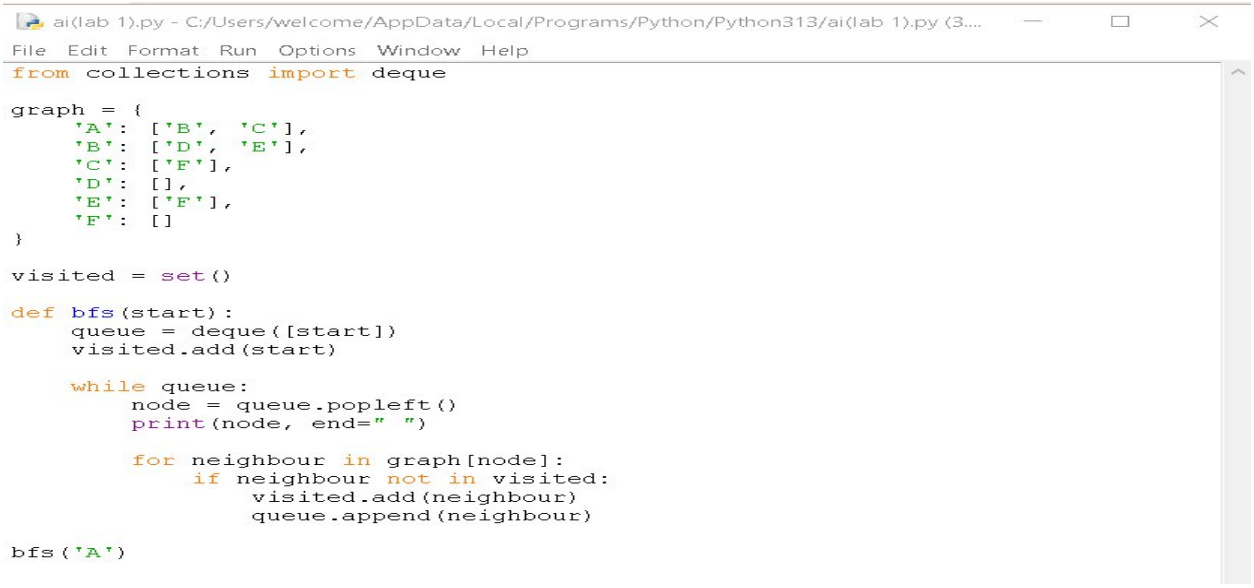
```
Starting Node:
A
```

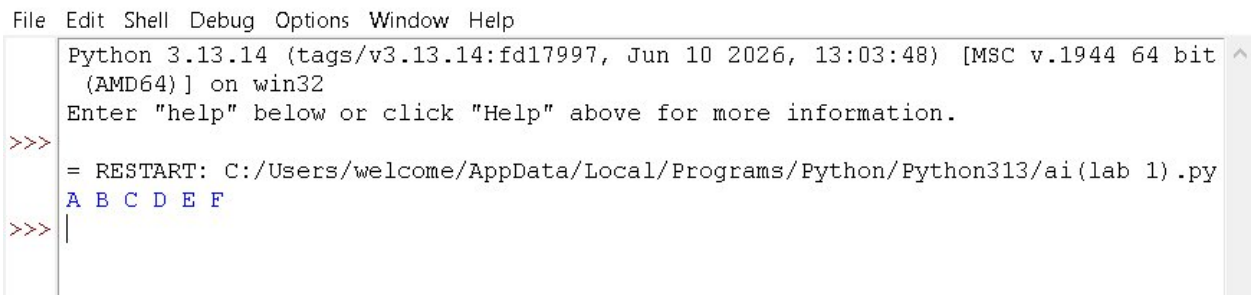
Output

```
A B C D E F
```

Result

Thus, the Breadth First Search (BFS) algorithm was implemented successfully.





8. Depth First Search (DFS)

Aim

To write a Python program to implement the Depth First Search (DFS) algorithm for graph traversal.

Algorithm

1. Start from the source node.
2. Mark the node as visited.
3. Display the node.
4. Recursively visit all unvisited adjacent nodes.
5. Continue until all nodes are visited.

Program

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            dfs(neighbour)

dfs('A')
```

Input

```
Starting Node:
A
```

Output

```
A B D E F C
```

Result

Thus, the Depth First Search (DFS) algorithm was implemented successfully.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

File Edit Format Run Options Window Help

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            dfs(neighbour)

dfs('A')
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
A B D E F C
>>>
```

9. Travelling Salesman Problem (TSP)

Aim

To write a Python program to solve the Travelling Salesman Problem using brute force.

Algorithm

1. Read the distance matrix.
2. Generate all possible city permutations.
3. Calculate the total distance for each route.
4. Select the route with the minimum distance.
5. Display the minimum cost.

Program

```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
cities = range(1, n)

min_cost = float('inf')

for path in permutations(cities):
    cost = 0
    k = 0

    for city in path:
        cost += graph[k][city]
        k = city

    cost += graph[k][0]

    if cost < min_cost:
        min_cost = cost

print("Minimum Cost:", min_cost)
```

Input

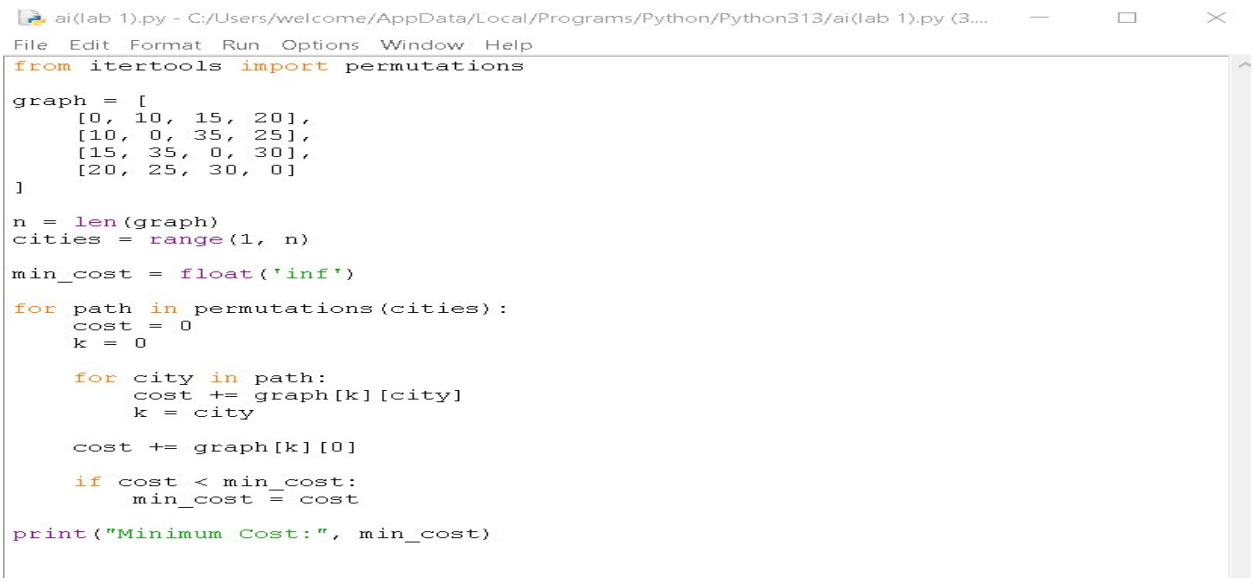
```
Distance Matrix:
0 10 15 20
10 0 35 25
15 35 0 30
20 25 30 0
```

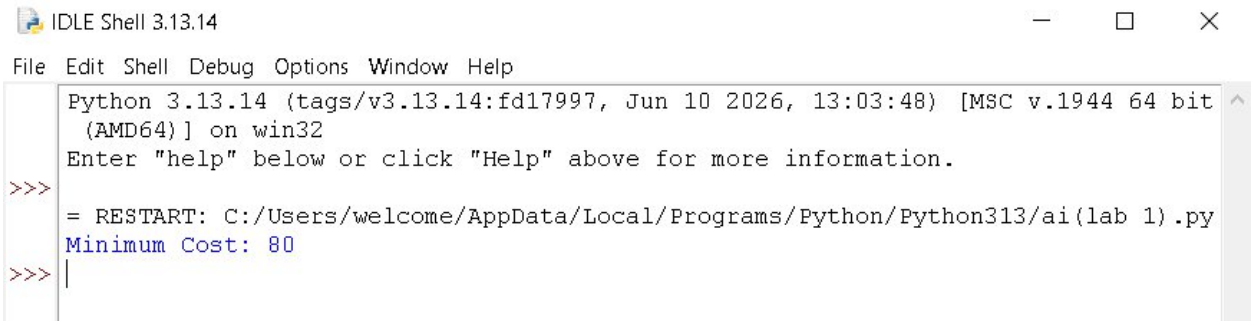
Output

```
Minimum Cost: 80
```

Result

Thus, the Travelling Salesman Problem was solved successfully using Python.





10. A (A-Star) Algorithm*

Aim

To write a Python program to implement the A* search algorithm.

Algorithm

1. Initialize the open list with the start node.
2. Maintain a closed list for visited nodes.
3. Select the node with the lowest **f = g + h** value.
4. If the goal node is reached, stop.
5. Otherwise, expand the node and update the costs.
6. Repeat until the goal is found.

Program

```
from queue import PriorityQueue

graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 3), ('E', 6)],
    'C': [('F', 5)],
    'D': [],
    'E': [('F', 2)],
    'F': []
}

heuristic = {
    'A': 6,
    'B': 4,
    'C': 4,
    'D': 2,
    'E': 1,
    'F': 0
}

def astar(start, goal):
    pq = PriorityQueue()
    pq.put((0, start))
    cost = {start: 0}

    while not pq.empty():
        _, node = pq.get()

        if node == goal:
            print("Goal Reached")
            return

        for neighbour, weight in graph[node]:
```

```
            new_cost = cost[node] + weight

            if neighbour not in cost or new_cost < cost[neighbour]:
                cost[neighbour] = new_cost
                priority = new_cost + heuristic[neighbour]
                pq.put((priority, neighbour))

astar('A', 'F')
```

Input

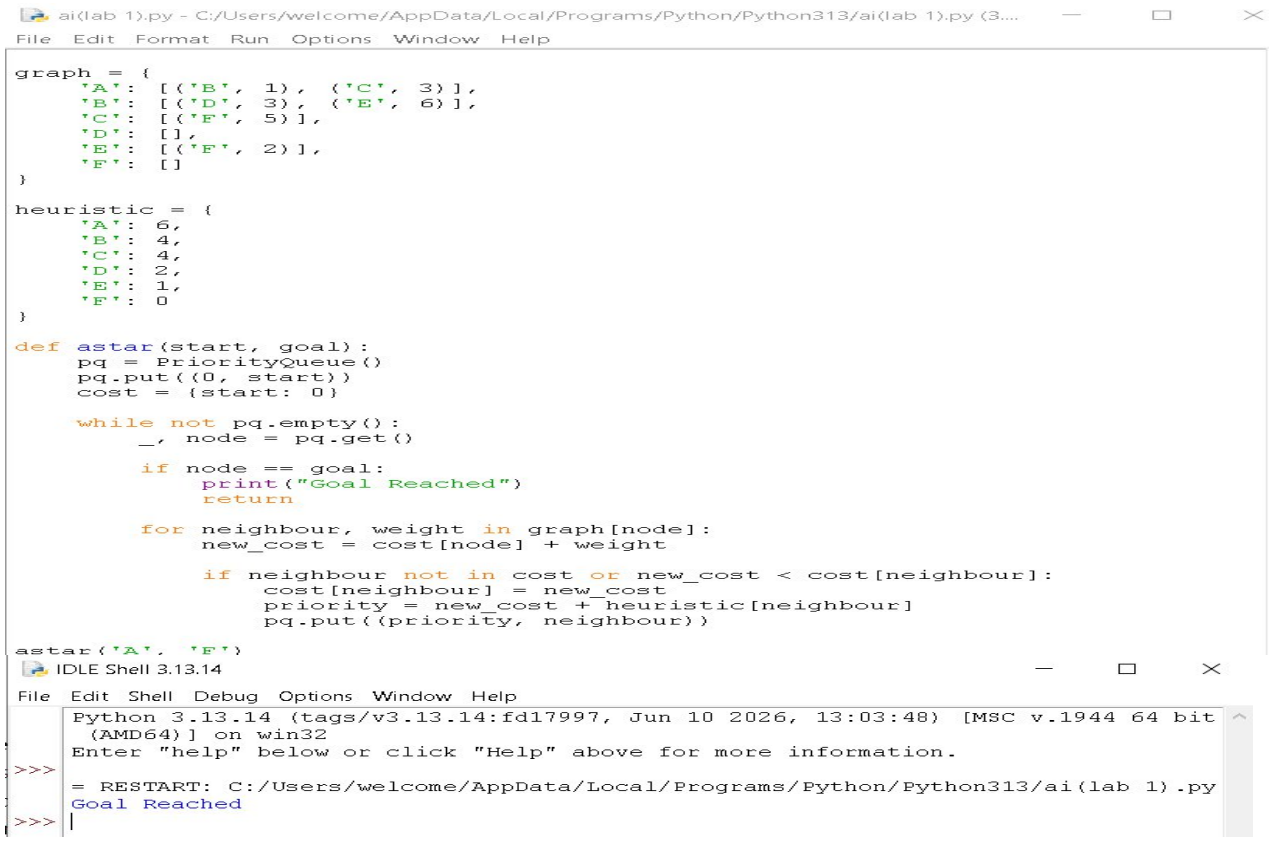
```
Start Node:
A

Goal Node:
F
```

Output

```
Goal Reached
```

Result

Thus, the A* search algorithm was implemented successfully in Python.



11. Map Coloring using CSP

Aim

To write a Python program to solve the **Map Coloring Problem** using **Constraint Satisfaction Problem (CSP)**.

Algorithm

1. Start with a set of regions and available colors.
2. Represent each region as a variable.
3. Assign a color to each region.
4. Check whether the assigned color conflicts with neighboring regions.
5. If there is no conflict, move to the next region.
6. If a conflict occurs, backtrack and try another color.
7. Repeat until all regions are colored.
8. Display the final color assignment.

Program

```
def map_coloring():
    regions = ['A', 'B', 'C', 'D']
    neighbors = {
        'A': ['B', 'C'],
        'B': ['A', 'C', 'D'],
        'C': ['A', 'B', 'D'],
        'D': ['B', 'C']
    }
    colors = ['Red', 'Green', 'Blue']
    assignment = {}
    def is_valid(region, color):
        for neighbor in neighbors[region]:
            if neighbor in assignment and assignment[neighbor] == color:
                return False
        return True
    def solve(index):
        if index == len(regions):
```

```
            return True
        region = regions[index]
        for color in colors:
            if is_valid(region, color):
                assignment[region] = color
                if solve(index + 1):
                    return True
                del assignment[region]
       return False
    if solve(0):
        print("Map Coloring Solution:")
        for region in regions:
            print(region, ":", assignment[region])
    else:
        print("No solution found")
map_coloring()
```

Input

Regions = A, B, C, D

Colors = Red, Green, Blue

Output

```
Map Coloring Solution:
A : Red
B : Green
C : Blue
D : Red
```

Result

Thus, the **Map Coloring Problem** was implemented successfully in Python using the **CSP backtracking algorithm**.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

File Edit Format Run Options Window Help

```
        'A': ['B', 'C'],
        'B': ['A', 'C', 'D'],
        'C': ['A', 'B', 'D'],
        'D': ['B', 'C']
    }

    colors = ['Red', 'Green', 'Blue']
    assignment = {}

    def is_valid(region, color):
        for neighbor in neighbors[region]:
            if neighbor in assignment and assignment[neighbor] == color:
                return False
        return True

    def solve(index):
        if index == len(regions):
            return True

        region = regions[index]

        for color in colors:
            if is_valid(region, color):
                assignment[region] = color

                if solve(index + 1):
                    return True

                del assignment[region]

        return False

    if solve(0):
        print("Map Coloring Solution:")
        for region in regions:
            print(region, ":", assignment[region])
    else:
        print("No solution found")

map_coloring()
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
Map Coloring Solution:
A : Red
B : Green
C : Blue
D : Red
>>>
```

12. Tic Tac Toe Game

Aim

To write a Python program to implement a simple **Tic Tac Toe game**.

Algorithm

1. Start with an empty 3 × 3 board.
2. Display the board.
3. Allow Player X to enter a position.

Program

```
def display(board):
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])
def check_winner(board, player):
    wins = [
        [0, 1, 2],
        [3, 4, 5],
        [6, 7, 8],
        [0, 3, 6],
        [1, 4, 7],
        [2, 5, 8],
        [0, 4, 8],
        [2, 4, 6]
    ]

    for w in wins:
        if all(board[i] == player for i in w):
            return True
    return False
```

```
board = ['1', '2', '3', '4', '5', '6', '7', '8', '9']
player = 'X'
for turn in range(9):
    display(board)
    pos = int(input("Enter position: ")) - 1
    if board[pos] in ['X', 'O']:
        print("Position already occupied")
        continue
    board[pos] = player
    if check_winner(board, player):
        display(board)
        print("Player", player, "wins")
        break
    player = 'O' if player == 'X' else 'X'
else:
    display(board)
    print("Game Draw")
```

Input

```
Enter position: 1
Enter position: 4
Enter position: 2
Enter position: 5
Enter position: 3
```

Output

```
X | X | X
--+---+--
O | O | 6
--+---+--
7 | 8 | 9


Player X wins
```

Result

Thus, the **Tic Tac Toe game** was implemented successfully in Python.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

File Edit Format Run Options Window Help

```
def display(board):
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])


def check_winner(board, player):
    wins = [
        [0, 1, 2],
        [3, 4, 5],
        [6, 7, 8],
        [0, 3, 6],
        [1, 4, 7],
        [2, 5, 8],
        [0, 4, 8],
        [2, 4, 6]
    ]

    for w in wins:
        if all(board[i] == player for i in w):
            return True

    return False


board = ['1', '2', '3', '4', '5', '6', '7', '8', '9']
player = 'X'

for turn in range(9):
    display(board)

    pos = int(input("Enter position: ")) - 1

    if board[pos] in ['X', 'O']:
        print("Position already occupied")
        continue

    board[pos] = player
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
1 | 2 | 3
--+---+--
4 | 5 | 6
--+---+--
7 | 8 | 9
Enter position:
```

13. Minimax Algorithm for Gaming

Aim

To write a Python program to implement the **Minimax Algorithm** for gaming.

Algorithm

1. Start with the current game state.
2. Generate all possible moves.
3. Check whether the state is a terminal state.
4. Assign a score to the terminal state.
5. The maximizing player selects the maximum score.
6. The minimizing player selects the minimum score.
7. Continue recursively until the best move is found.
8. Display the best move.

Program

```
def minimax(depth, is_max):
    scores = [10, 5, -10, 0]
    if depth == 0:
        return scores[0]
    if is_max:
        best = -1000
        for i in range(2):
            value = minimax(depth - 1, False)
            best = max(best, value)
        return best
    else:
        best = 1000
        for i in range(2):
            value = minimax(depth - 1, True)
            best = min(best, value)
        return best


print("Best Score:", minimax(3, True))
```

Input

```
Game depth = 3
```

Output

```
Best Score: 10
```

Result

Thus, the **Minimax Algorithm** was implemented successfully in Python for gaming.

ai(lab 1).py - C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py (3....

File Edit Format Run Options Window Help

```
def minimax(depth, is_max):
    scores = [10, 5, -10, 0]

    if depth == 0:
        return scores[0]

    if is_max:
        best = -1000

        for i in range(2):
            value = minimax(depth - 1, False)
            best = max(best, value)

        return best

    else:
        best = 1000

        for i in range(2):
            value = minimax(depth - 1, True)
            best = min(best, value)

        return best


print("Best Score:", minimax(3, True))
```

```
= RESTART: C:/Users/welcome/AppData/Local/Programs/Python/Python313/ai(lab 1).py
Best Score: 10
>>>
```

14. Alpha-Beta Pruning Algorithm for Gaming

Aim

To write a Python program to implement the **Alpha-Beta Pruning Algorithm** for gaming.

Algorithm

1. Start with the game tree.
2. Set alpha = -∞ and beta = +∞.
3. Generate possible moves.
4. Find the maximum value for the maximizing player.
5. Find the minimum value for the minimizing player.
6. Update alpha and beta values.
7. If alpha becomes greater than or equal to beta, prune the remaining branches.
8. Continue until the best move is obtained.
9. Display the best score.

Program

```
def alphabeta(depth, alpha, beta, is_max):
    if depth == 0:
        return 10
    if is_max:
        best = -1000
        for i in range(2):
            value = alphabeta(depth - 1, alpha, beta, False)
            best = max(best, value)
            alpha = max(alpha, best)
            if alpha >= beta:
                break
        return best
    else:
        best = 1000


        for i in range(2):
```

```
            value = alphabeta(depth - 1, alpha, beta, True)
            best = min(best, value)
            beta = min(beta, best)
            if alpha >= beta:
                break
        return best
print("Best Score:",
      alphabeta(3, -1000, 1000, True))
```

Input

```
Game depth = 3
Alpha = -1000
Beta = 1000
```

Output

```
Best Score: 10
```

Result

Thus, the **Alpha-Beta Pruning Algorithm** was implemented successfully in Python for gaming.

15. Decision Tree

Aim

To write a Python program to implement a **Decision Tree** for classification.

Algorithm

9. Import the required libraries.
10. Create the input dataset.
11. Separate the features and target values.
12. Create a Decision Tree Classifier.
13. Train the model using the dataset.
14. Give a new input value to the model.
15. Predict the class.
16. Display the predicted result.

Program

```
from sklearn.tree import DecisionTreeClassifier

# Input data

X = [

    [20, 1],

    [25, 1],

    [30, 0],

    [35, 0],

    [40, 0],

    [45, 1]

]

# Target data

y = [0, 0, 1, 1, 1, 1]

# Create Decision Tree

model = DecisionTreeClassifier()

# Train the model

model.fit(X, y)

# Predict for new data

age = int(input("Enter age: "))
```

```
family_history = int(input("Family history (1-Yes, 0-No): "))

prediction = model.predict([[age, family_history]])

print("Predicted Class:", prediction[0])
```

Input

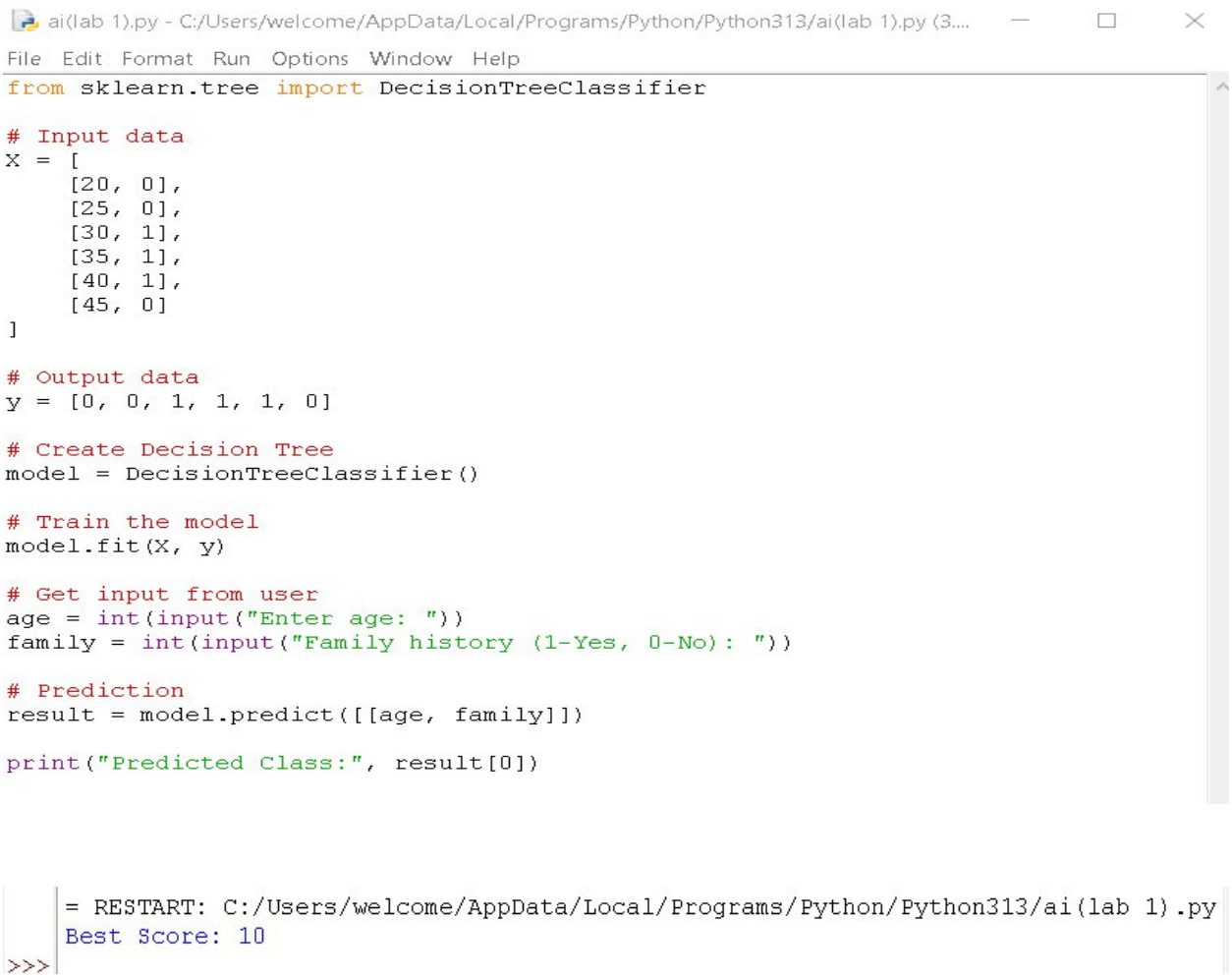
```
Enter age: 30

Family history (1-Yes, 0-No): 0
```

Output

```
Predicted Class: 1
```

Result

Thus, the **Decision Tree classification algorithm** was implemented successfully in Python.



16. Feed Forward Neural Network

Aim

To write a Python program to implement a **Feed Forward Neural Network** for classification.

Algorithm

17. Import the required libraries.
18. Create the input dataset and target values.
19. Create a Feed Forward Neural Network using MLPClassifier.
20. Train the network using the training data.
21. Give a new input to the trained network.
22. Predict the output class.
23. Display the predicted result.

Program

```
from sklearn.neural_network import MLPClassifier

# Input data

X = [

    [0, 0],

    [0, 1],

    [1, 0],

    [1, 1]

]

# Target data

y = [0, 1, 1, 0]

# Create Feed Forward Neural Network

model = MLPClassifier(

    hidden_layer_sizes=(4,),

    max_iter=2000,

    random_state=1

)

# Train the network

model.fit(X, y)
```

```
# Get input

a = int(input("Enter first value (0 or 1): "))

b = int(input("Enter second value (0 or 1): "))

# Predict output

prediction = model.predict([[a, b]])

print("Predicted Output:", prediction[0])
```

Input

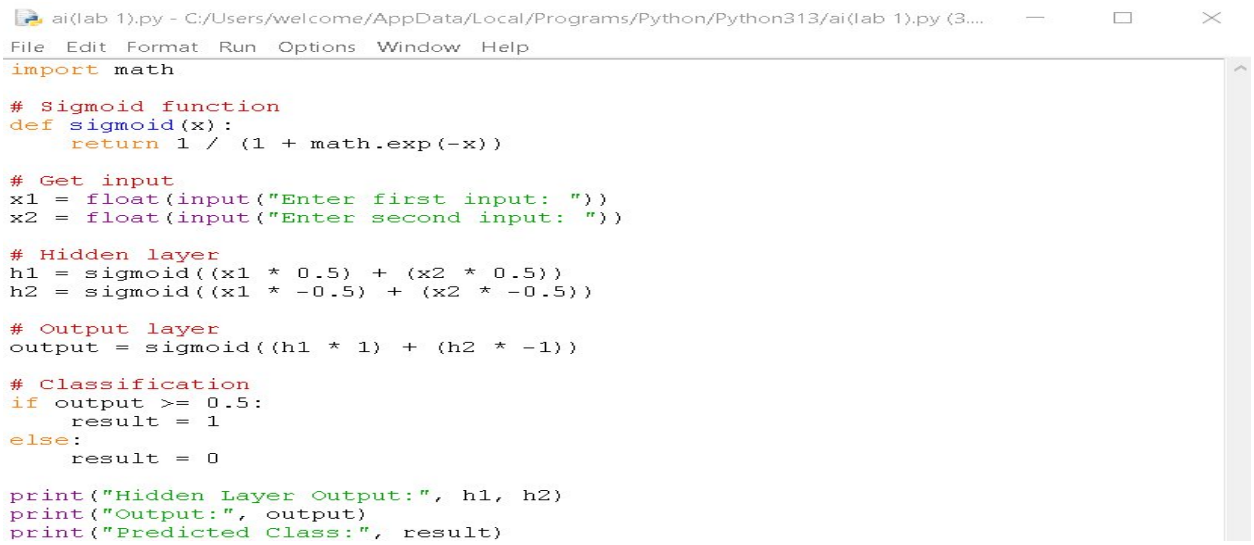
```
Enter first value (0 or 1): 1

Enter second value (0 or 1): 0
```

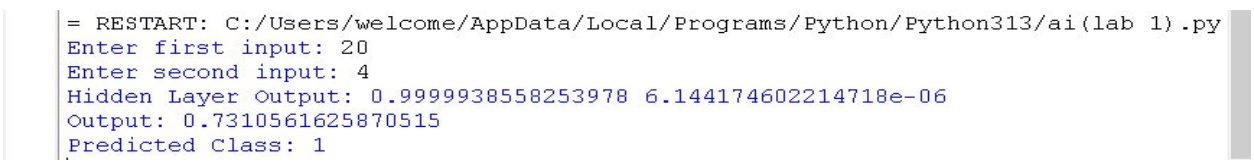
Output

```
Predicted Output: 1
```

Result

Thus, the **Feed Forward Neural Network** was implemented successfully in Python for classification.





17: Sum of Integers from 1 to N

Aim

To write and execute a Prolog program to find the sum of integers from 1 to N.

Algorithm

1. Start the program.
2. Define the base case sum(0,0).
3. Check whether N is greater than 0.
4. Decrease N by 1.
5. Recursively calculate the sum.
6. Add N to the recursive sum.
7. Display the result.
8. Stop.

Program

```
sum(0, 0).

sum(N, S) :-

  N > 0,

  N1 is N - 1,

  sum(N1, S1),

  S is N + S1.
```

Output

```
?- sum(5, S).

S = 15.
```

Result

Thus, the Prolog program was successfully executed to calculate the sum of integers from 1 to N.

18: Prolog Program for Database with NAME and DOB

Aim

To write and execute a Prolog program to create a database containing names and dates of birth and retrieve the required information.

Algorithm

1. Start the program.
2. Define facts containing a person's name and date of birth.
3. Store the facts using the predicate person(Name, DOB).
4. Use a query to search for a person's DOB.
5. Display the matching result.
6. Stop.

Program

```
person(ragul, '10-05-2005').

person(rahul, '15-08-2004').

person(anu, '20-12-2005').

person(kavin, '05-03-2004').


find_dob(Name, DOB) :-

   person(Name, DOB).
```

Output

```
?- find_dob(ragul, DOB).

DOB = '10-05-2005'.


?- find_dob(anu, DOB).

DOB = '20-12-2005'.
```

Result

Thus, the Prolog database was successfully created and the date of birth was retrieved using queries.


SWI-Prolog -- C:/Users/welcome/Desktop/experiment18.pl
File Settings Tools Debug GUI Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.1.13)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
19 ?- [experiment18].
true.
20 ?- person(ragul, DOB).
DOB = '10-05-2005'.
21 ?- person(anu, DOB).
DOB = '20-12-2005'.

19: Prolog Program for Student-Teacher-Subject Code

Aim

To write and execute a Prolog program to represent the relationship between students, teachers, and subject codes.

Algorithm

1. Start the program.
2. Define facts for students and their subject codes.
3. Define facts for teachers and the subjects they handle.
4. Create a rule to find the teacher for a student's subject.
5. Execute the required query.
6. Display the result.
7. Stop.

Program

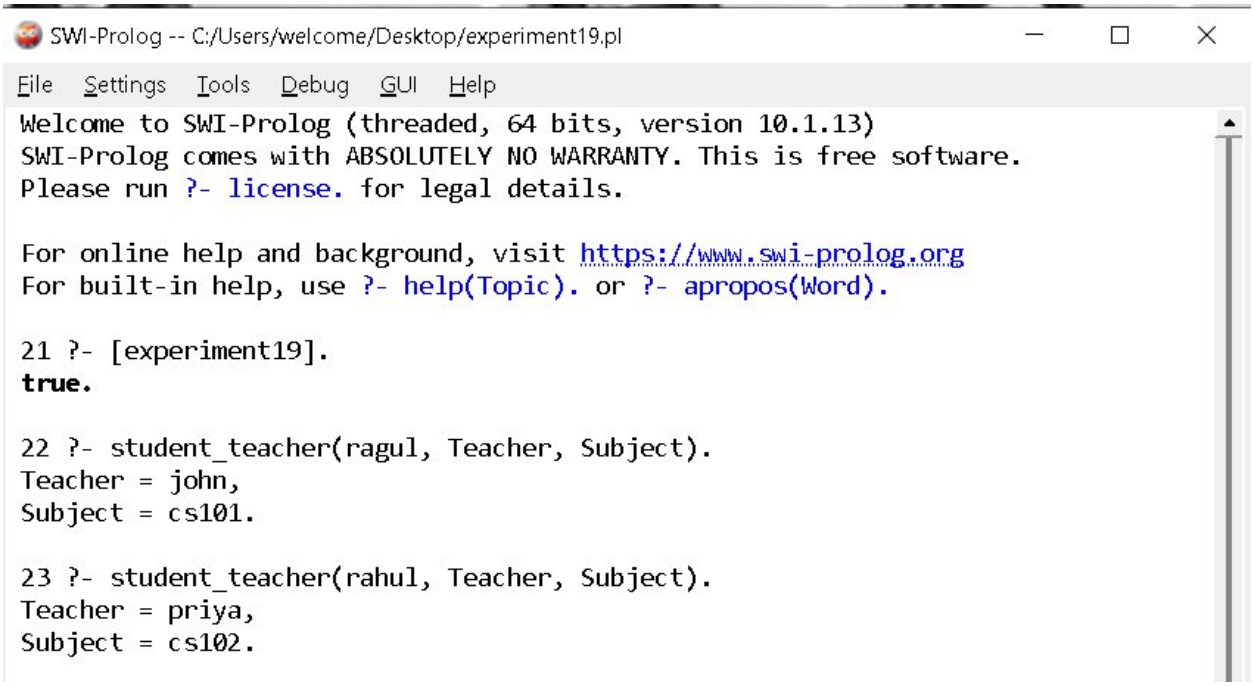
```
student(ragul, cs101).

student(rahul, cs102).

student(anu, cs103).



teacher(john, cs101).

teacher(priya, cs102).

teacher(kumar, cs103).



student_teacher(Student, Teacher, Subject) :-

   student(Student, Subject),

   teacher(Teacher, Subject).
```

Output

```
?- student_teacher(ragul, Teacher, Subject).

Teacher = john,

Subject = cs101.

?- student_teacher(rahul, Teacher, Subject).

Teacher = priya,

Subject = cs102.
```

Result

Thus, the Prolog program was successfully executed to find the teacher and subject code associated with a student.



20: Prolog Program for Planets Database

Aim

To write and execute a Prolog program to create a database of planets and retrieve planet information.

Algorithm

1. Start the program.
2. Define facts for the planets.
3. Store the planet names using the planet predicate.
4. Use a query to check whether a planet exists.
5. Display the result.
6. Stop.

Program

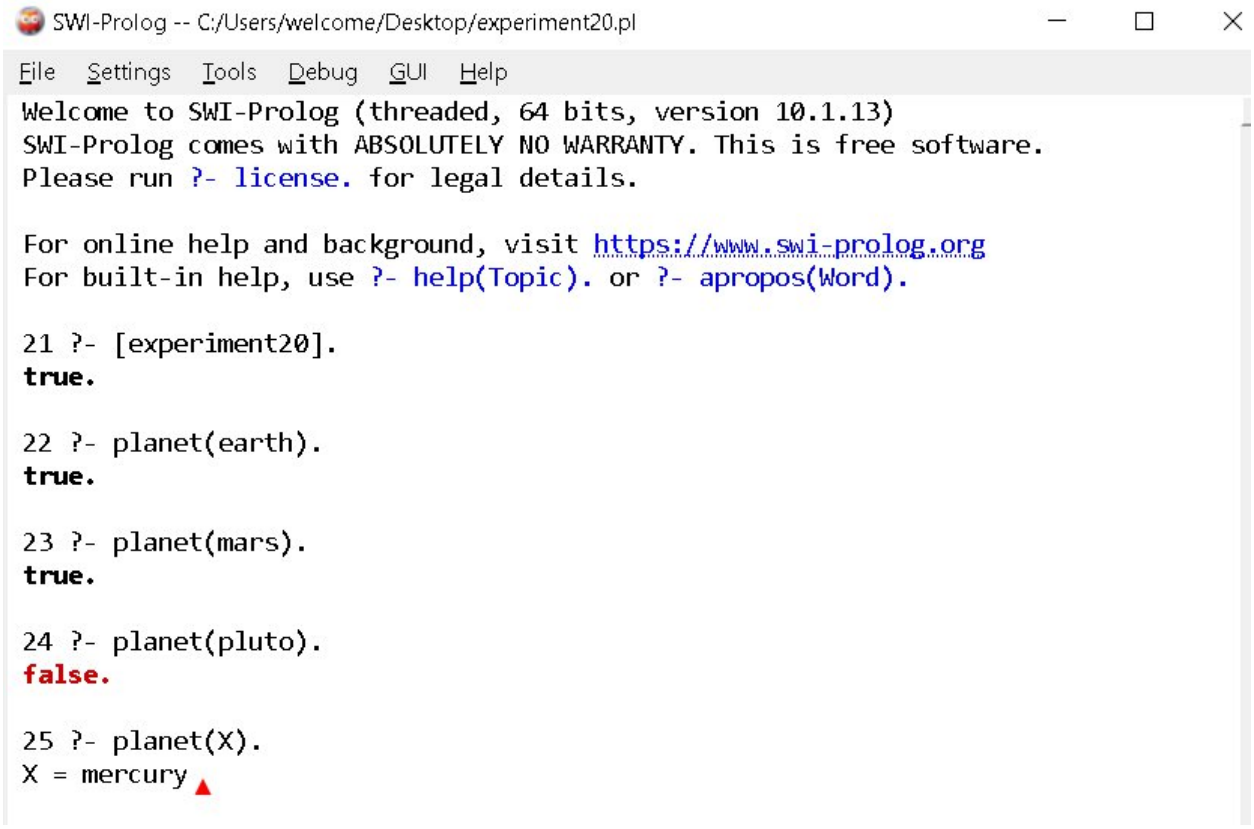
```
planet(mercury).

planet(venus).

planet(earth).

planet(mars).

planet(jupiter).

planet(saturn).

planet(uranus).

planet(neptune).


is_planet(X) :-

   planet(X).
```

Output

```
?- is_planet(earth).

true.


?- is_planet(mars).

true.


?- is_planet(pluto).

false.
```

Result

Thus, the Prolog program was successfully executed to create and query the planets database.



21: Prolog Program to Implement Towers of Hanoi

Aim

To write and execute a Prolog program to solve the **Towers of Hanoi** problem using recursion.

Algorithm

1. Start the program.
2. If there is only one disk, move it directly from source to destination.
3. For N disks, move N-1 disks from source to auxiliary peg.
4. Move the largest disk from source to destination.
5. Move N-1 disks from auxiliary peg to destination.
6. Repeat recursively until all disks are moved.
7. Display each move.
8. Stop.

Program

```
hanoi(1, Source, Destination, _) :-
    write('Move disk from '),
    write(Source),
    write(' to '),
    write(Destination),
    nl.


hanoi(N, Source, Destination, Auxiliary) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, Source, Auxiliary, Destination),
    hanoi(1, Source, Destination, Auxiliary),
    hanoi(N1, Auxiliary, Destination, Source).
```

Output

Query:

```
?- hanoi(3, left, right, middle).
```

Output:

```
Move disk from left to right
```

Move disk from left to middle

Move disk from right to middle

Move disk from left to right

Move disk from middle to left

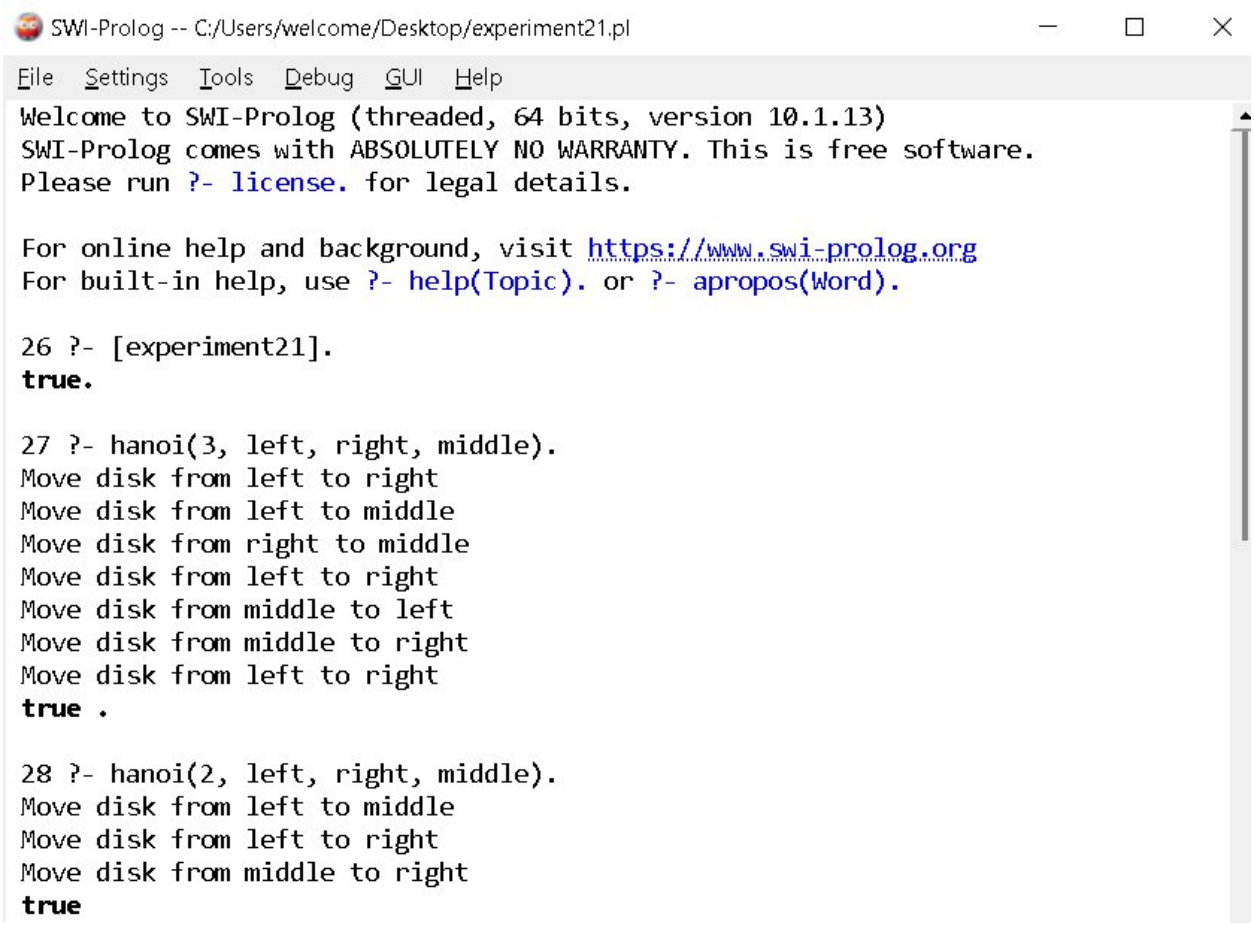
Move disk from middle to right

Move disk from left to right

true.

Result

Thus, the Prolog program was successfully executed to implement the **Towers of Hanoi** using recursion.



21: Towers of Hanoi

Aim

To write and execute a Prolog program to implement the Towers of Hanoi problem using recursion.

Algorithm

1. Start the program.
2. If there is one disk, move it directly to the destination.
3. Move N−1 disks from source to auxiliary.
4. Move the remaining disk from source to destination.
5. Move N−1 disks from auxiliary to destination.
6. Repeat recursively until all disks are moved.
7. Stop.

Program

```
hanoi(1, Source, Destination, _) :-
   write('Move disk from '),
   write(Source),
   write(' to '),
   write(Destination),
   nl.


hanoi(N, Source, Destination, Auxiliary) :-
   N > 1,
   N1 is N - 1,
   hanoi(N1, Source, Auxiliary, Destination),
   hanoi(1, Source, Destination, Auxiliary),
   hanoi(N1, Auxiliary, Destination, Source).
```

Output

```
?- hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
```

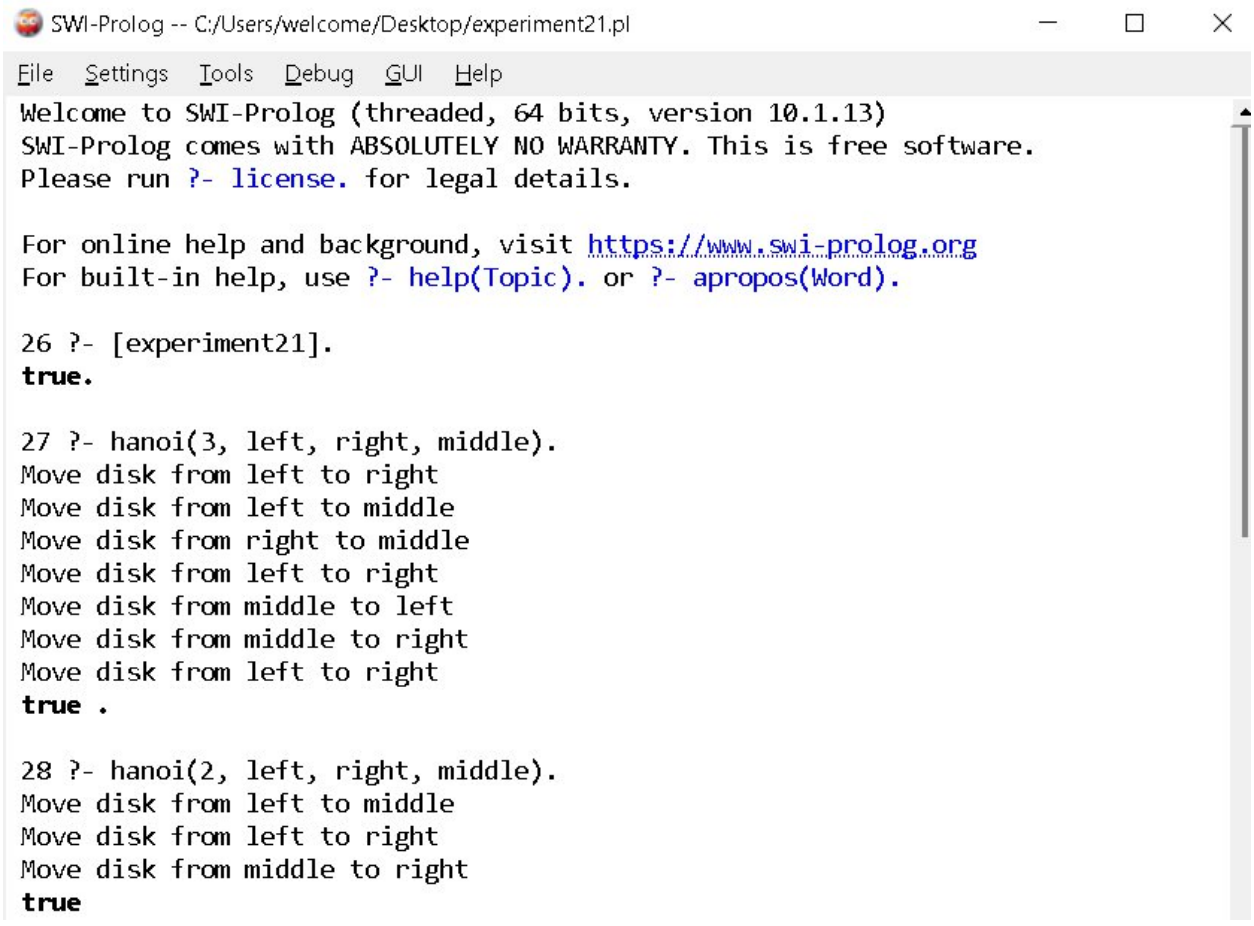
Move disk from middle to right

Move disk from left to right

true.

Result

Thus, the Prolog program was successfully executed to implement the Towers of Hanoi problem.



22: Prolog Program to Check Whether a Bird Can Fly

Aim

To write and execute a Prolog program to determine whether a particular bird can fly using facts and rules.

Algorithm

1. Start the program.
2. Define facts for birds.
3. Define birds that cannot fly.
4. Create a rule to determine whether a bird can fly.
5. Give the required query.
6. Display the result.
7. Stop.

Program

```
bird(parrot).

bird(eagle).

bird(pigeon).

bird(ostrich).

bird(penguin).


cannot_fly(ostrich).

cannot_fly(penguin).


can_fly(Bird) :-

   bird(Bird),

   \+ cannot_fly(Bird).
```

Output

```
?- can_fly(parrot).

true.


?- can_fly(eagle).

true.
```

?- can_fly(penguin).

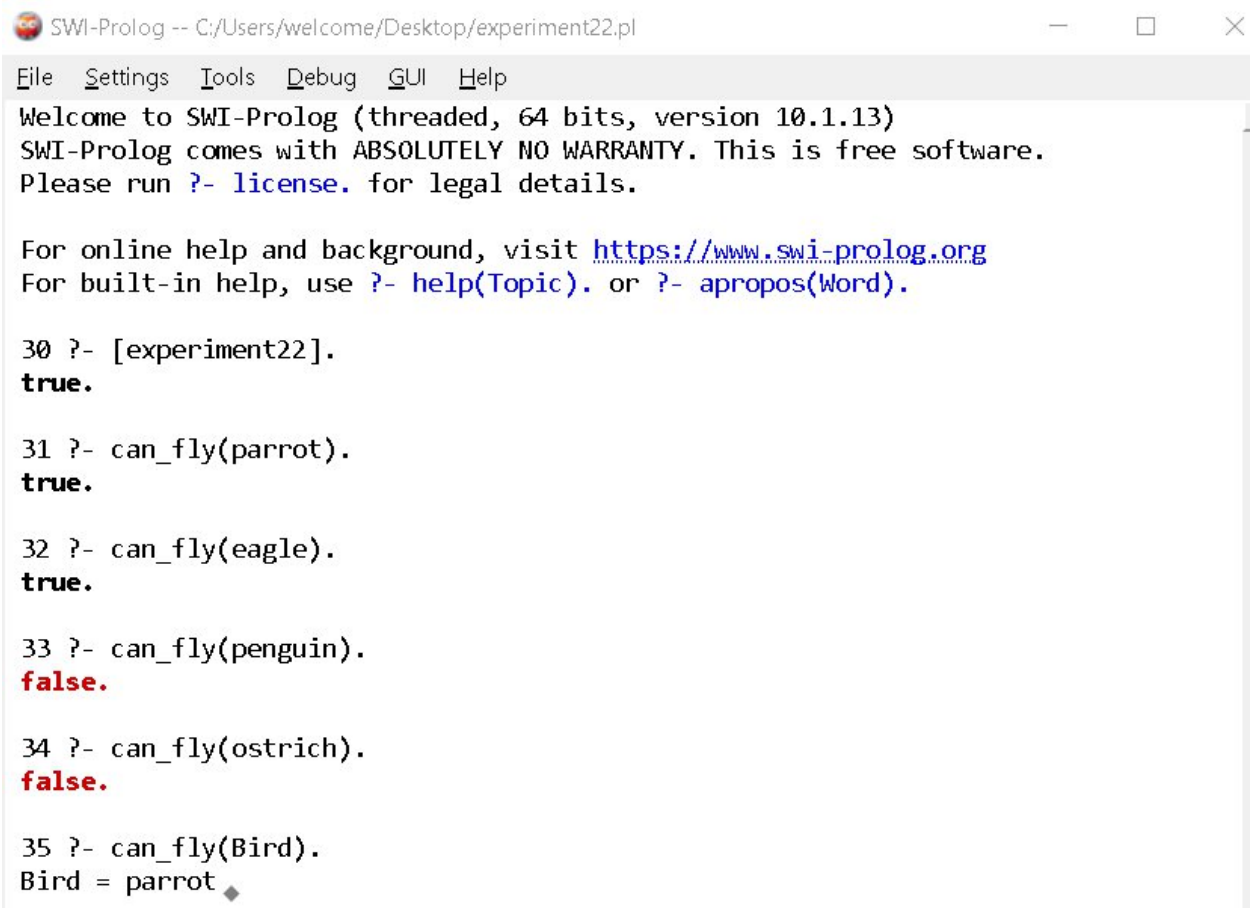
false.

?- can_fly(ostrich).

false.

Result

Thus, the Prolog program was successfully executed to determine whether a particular bird can fly or not.



23: Prolog Program to Implement Family Tree

Aim

To write and execute a Prolog program to represent a family tree and find relationships between family members.

Algorithm

1. Start the program.
2. Define facts for parent relationships.
3. Define rules for father, mother, grandparent, and sibling relationships.
4. Use queries to find relationships.
5. Display the results.
6. Stop.

Program

```
parent(john, mary).

parent(john, david).

parent(susan, mary).

parent(susan, david).

parent(david, tom).

parent(david, anna).

father(X, Y) :-

   parent(X, Y),

   male(X).

mother(X, Y) :-

   parent(X, Y),

   female(X).

male(john).

male(david).

male(tom).


female(susan).

female(mary).

female(anna).
```

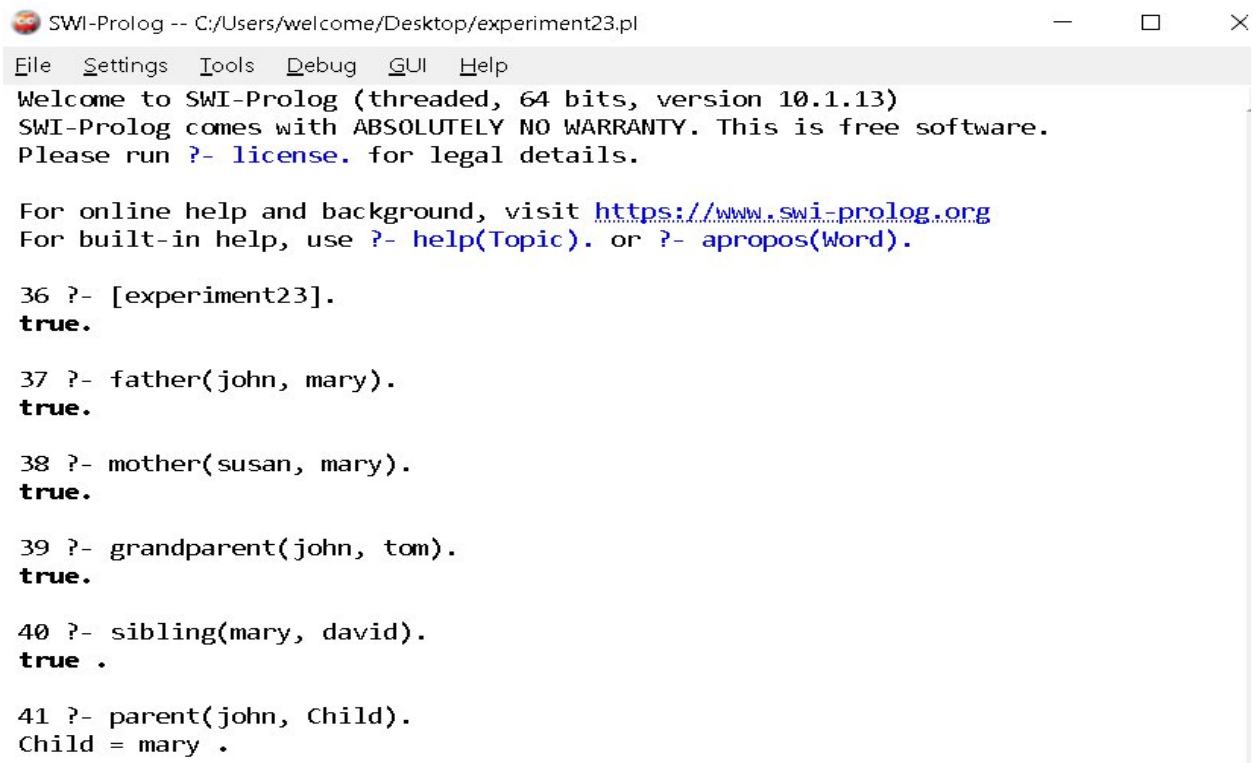
```
grandparent(X, Z) :-
    parent(X, Y),
    parent(Y, Z).
sibling(X, Y) :-
    parent(P, X),
    parent(P, Y),
    X \= Y.
```

Output

```
?- grandparent(john, tom).
true.
?- sibling(mary, david).
true.
?- parent(david, anna).
true.
```

Result

Thus, the Prolog program was successfully executed to implement a family tree and determine family relationships.



24: Prolog Program for Dieting System Based on Disease

Aim

To write and execute a Prolog program to suggest suitable foods based on a person's disease.

Algorithm

1. Start the program.
2. Define diseases and recommended foods.
3. Define rules to suggest food according to disease.
4. Give the disease as input.
5. Retrieve the recommended food.
6. Display the result.
7. Stop.

Program

```
food(diabetes, vegetables).

food(diabetes, whole_grains).

food(diabetes, fruits).

food(hypertension, vegetables).

food(hypertension, banana).

food(hypertension, oats).

food(anemia, spinach).

food(anemia, beans).

food(anemia, apple).

suggest_food(Disease, Food) :-

   food(Disease, Food).
```

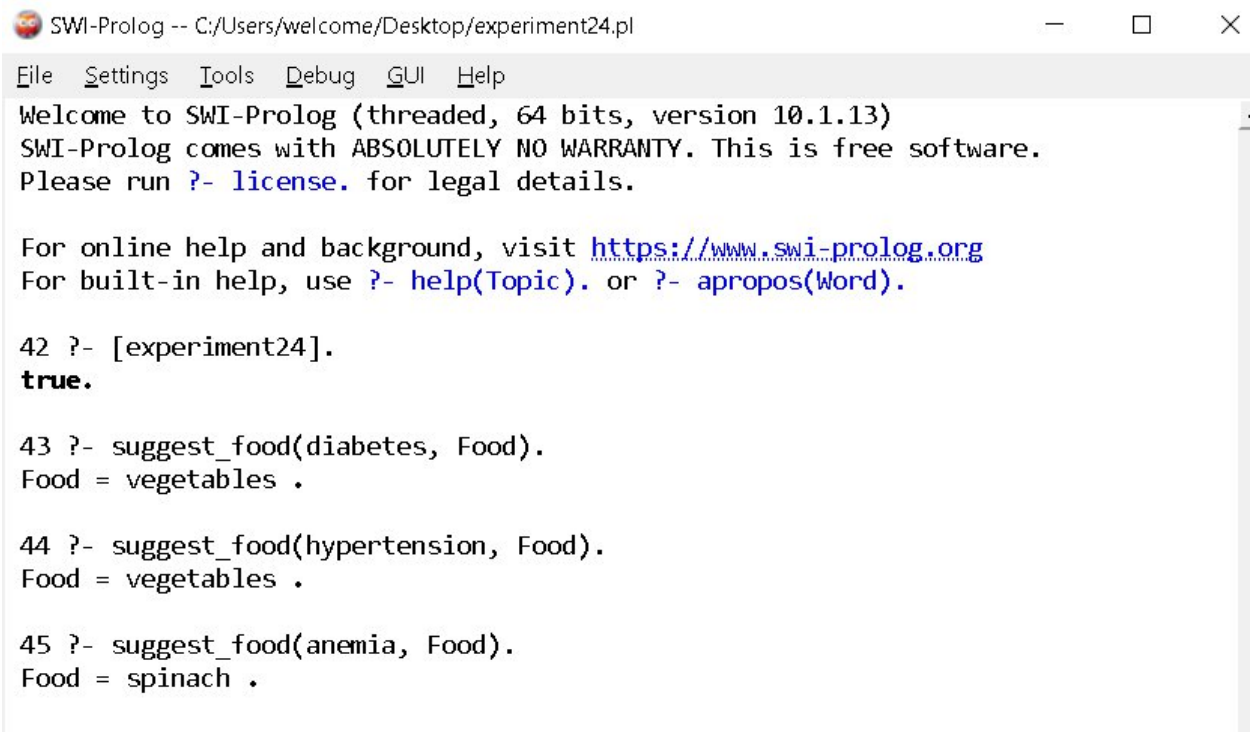
Output

```
?- suggest_food(diabetes, Food).

Food = vegetables ;

Food = whole_grains ;

Food = fruits.

?- suggest_food(anemia, Food).

Food = spinach ;

Food = beans ;
```

Food = apple.

Result

Thus, the Prolog program was successfully executed to suggest foods based on the given disease.



25: Prolog Program to Implement Monkey Banana Problem

Aim

To write and execute a Prolog program to solve the Monkey Banana problem using logical reasoning and state transitions.

Algorithm

1. Start the program.
2. Represent the monkey, box, and banana positions as states.
3. Check whether the monkey is under the banana and has the box.
4. If the box is not under the banana, move the box.
5. Push the box to the banana position.
6. Climb onto the box.
7. Grasp the banana.
8. Display the solution.
9. Stop.

Program

```
move(state(Monkey, Box, Banana, OnBox),

     state(Banana, Box, Banana, OnBox),

     move_monkey) :-

    Monkey \= Banana.


move(state(Monkey, Monkey, Banana, OnBox),

     state(Banana, Banana, Banana, OnBox),

     push_box) :-

    Monkey \= Banana.


move(state(Monkey, Box, Banana, no),

     state(Box, Box, Banana, yes),

     climb) :-

    Monkey = Box.


solve(state(Monkey, Box, Banana, no), [Action]) :-

    Monkey = Banana,
```

```
    Box = Banana,
    Action = grasp_banana.

solve(State, [Action|Actions]) :-
    move(State, NewState, Action),
    solve(NewState, Actions).
```

Output

?- solve(state(door, middle, middle, no), Actions).

Actions = [move_monkey, push_box, climb, grasp_banana].

Result

Thus, the Prolog program was successfully executed to implement the Monkey Banana problem using state-space reasoning.

26: Prolog Program for Fruit and Its Color Using Backtracking

Aim

To write and execute a Prolog program to identify the color of fruits using backtracking.

Algorithm

1. Start the program.
2. Define facts containing fruits and their colors.
3. Create a rule to identify the color of a fruit.
4. Give a fruit as input.
5. Prolog searches the database for matching facts.
6. Use backtracking to find additional solutions when available.
7. Display the result.
8. Stop.

Program

```
fruit_color(apple, red).

fruit_color(apple, green).

fruit_color(banana, yellow).

fruit_color(orange, orange).

fruit_color(grapes, green).

fruit_color(grapes, purple).

fruit_color(mango, yellow).


fruit_color_query(Fruit, Color) :-

   fruit_color(Fruit, Color).
```

Output

```
?- fruit_color_query(apple, Color).


Color = red ;

Color = green ;

false.
```

Another query:

```
?- fruit_color_query(grapes, Color).
```

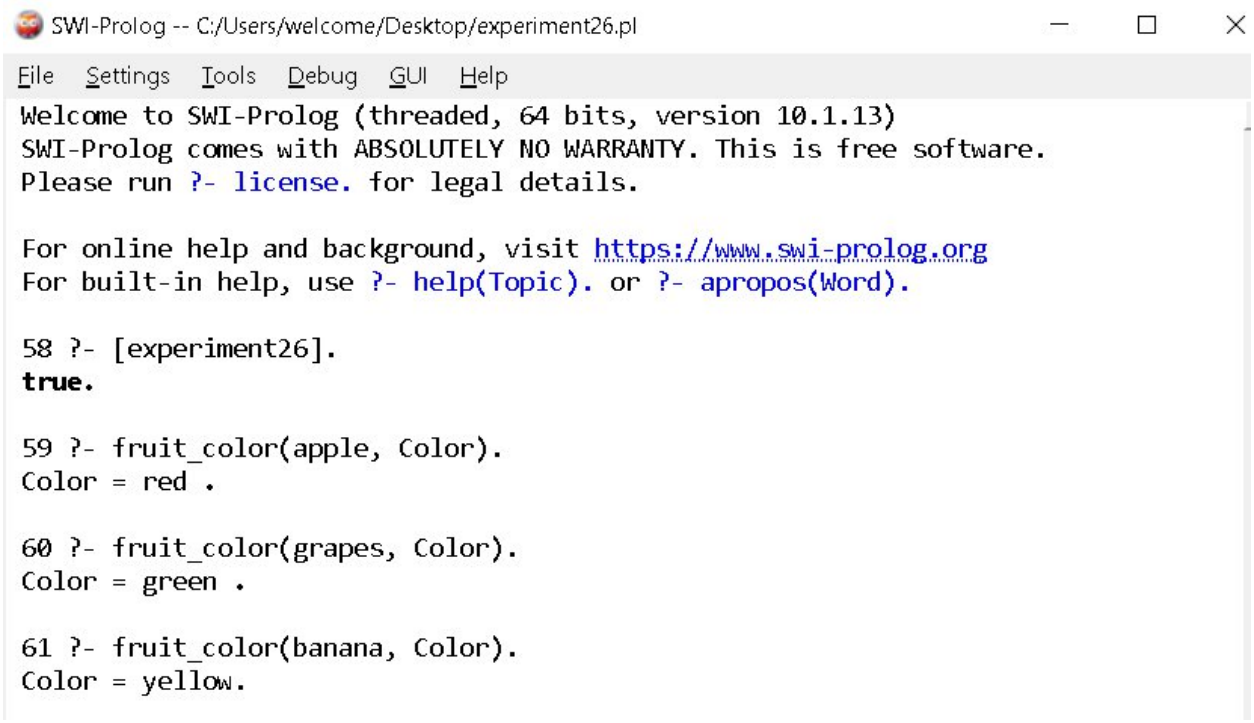
Color = green ;

Color = purple ;

false.

Result

Thus, the Prolog program was successfully executed to identify fruit colors using **backtracking**



27: Prolog Program to Implement Best First Search Algorithm

Aim

To write and execute a Prolog program to implement the Best First Search algorithm.

Algorithm

1. Start the program.
2. Represent the graph using nodes and heuristic values.
3. Select the node with the lowest heuristic value.
4. Expand the selected node.
5. Add its neighboring nodes to the open list.
6. Continue selecting the node with the best heuristic value.
7. Stop when the goal node is reached.
8. Display the path.

Program

```
edge(a, b).

edge(a, c).

edge(b, d).

edge(b, e).

edge(c, f).

edge(c, g).

edge(d, h).

edge(e, h).

edge(f, h).

edge(g, h).


heuristic(a, 10).

heuristic(b, 6).

heuristic(c, 5).

heuristic(d, 4).

heuristic(e, 3).

heuristic(f, 2).

heuristic(g, 1).
```

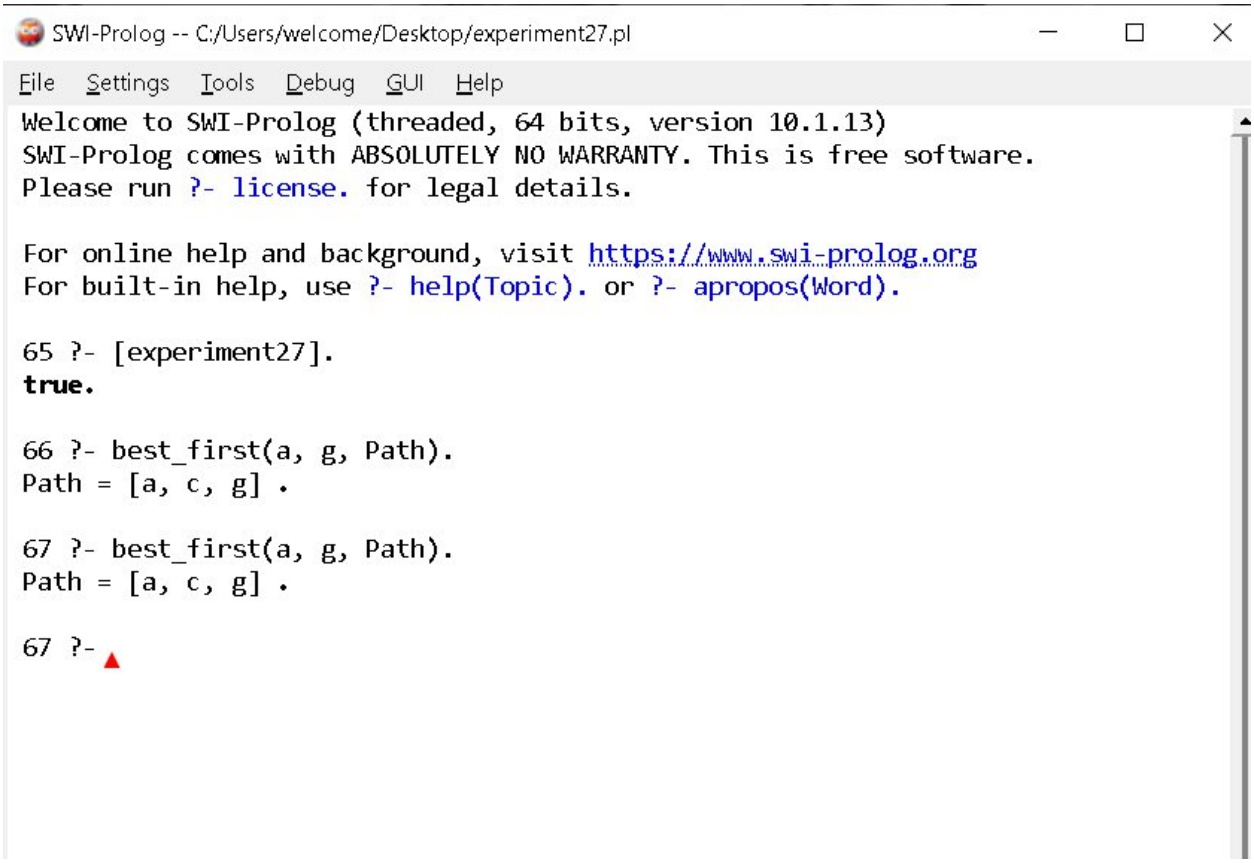
```
heuristic(h, 0).


best_first(Start, Goal, Path) :-
    search([Start], Goal, [Start], RevPath),
    reverse(RevPath, Path).


search([Goal|_], Goal, _, [Goal]).


search([Current|Open], Goal, Visited, [Current|Path]) :-
    findall(H-Next,
            (edge(Current, Next),
             \+ member(Next, Visited),
             heuristic(Next, H)),
            Children),
    keysort(Children, Sorted),
    pairs_values(Sorted, Nodes),
    append(Nodes, Open, NewOpen),
    NewOpen = [Next|_],
    search(NewOpen, Goal, [Next|Visited], Path).
```

Output

```
?- best_first(a, h, Path).


Path = [a, c, f, h].
```

Result

Thus, the Prolog program was successfully executed to implement the **Best First Search algorithm**.


SWI-Prolog -- C:/Users/welcome/Desktop/experiment27.pl
File Settings Tools Debug GUI Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.1.13)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
65 ?- [experiment27].
true.
66 ?- best_first(a, g, Path).
Path = [a, c, g] .
67 ?- best_first(a, g, Path).
Path = [a, c, g] .
67 ?-

28: Prolog Program for Medical Diagnosis

Aim

To write and execute a Prolog program to diagnose a disease based on the symptoms provided.

Algorithm

1. Start the program.
2. Define symptoms associated with different diseases.
3. Accept symptoms as input.
4. Compare the given symptoms with the stored facts.
5. Identify the matching disease.
6. Display the diagnosis.
7. Stop.

Program

```
symptom(fever, flu).

symptom(cough, flu).

symptom(body_pain, flu).


symptom(fever, malaria).

symptom(chills, malaria).

symptom(headache, malaria).


symptom(cough, cold).

symptom(sneezing, cold).

symptom(runny_nose, cold).


diagnosis(Disease, Symptom) :-

    symptom(Symptom, Disease).
```

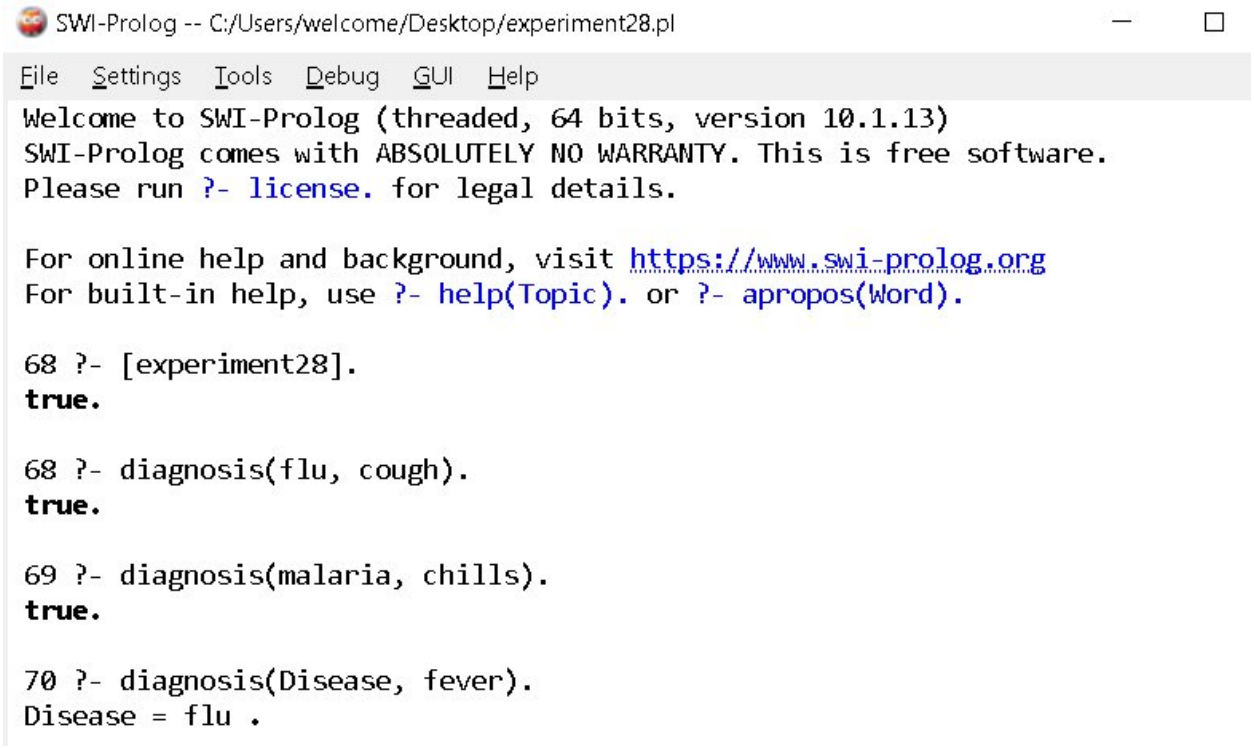
Output

```
?- diagnosis(flu, cough).

true.

?- diagnosis(malaria, chills).

true.

?- diagnosis(cold, fever).
```

```
false.
```

Result

Thus, the Prolog program was successfully executed to identify possible diseases based on the given symptoms.



29: Prolog Program for Forward Chaining

Aim

To write and execute a Prolog program to implement forward chaining using facts and rules.

Algorithm

1. Start the program.
2. Define the initial facts.
3. Define rules based on the facts.
4. Check the available facts.
5. Apply rules whose conditions are satisfied.
6. Derive new facts.
7. Continue until the required conclusion is obtained.
8. Display the result.

Program

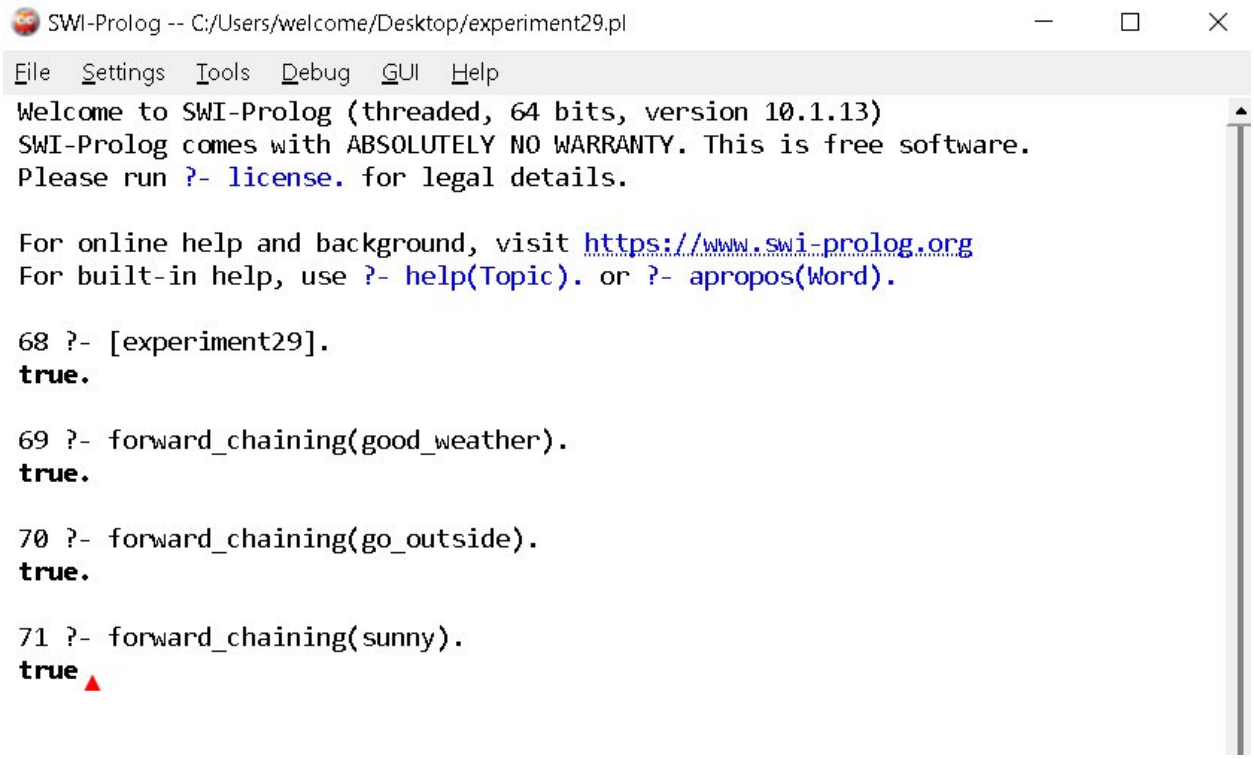
```
fact(sunny).

fact(warm).


rule(good_weather) :-

    fact(sunny),

    fact(warm).


rule(go_outside) :-

    rule(good_weather).


forward_chaining(X) :-

    fact(X).


forward_chaining(X) :-

    rule(X).
```

Output

```
?- forward_chaining(good_weather).


true.


?- forward_chaining(go_outside).


true.
```

Result

Thus, the Prolog program was successfully executed to demonstrate **Forward Chaining**.



30: Prolog Program for Backward Chaining

Aim

To write and execute a Prolog program to implement backward chaining using facts and rules.

Algorithm

1. Start the program.
2. Define facts and rules.
3. Set a goal to be proved.
4. Search for a rule whose conclusion matches the goal.
5. Treat the rule conditions as sub-goals.
6. Check whether the sub-goals can be proved using available facts or rules.
7. If all sub-goals are satisfied, prove the original goal.
8. Display the result.

Program

```
fact(sunny).

fact(warm).


good_weather :-

    fact(sunny),

    fact(warm).


go_outside :-

    good_weather.
```

Output

```
?- good_weather.


true.


?- go_outside.

true.

?- rainy.


false.
```

Result

Thus, the Prolog program was successfully executed to demonstrate **Backward Chaining**.

31: Create a Web Blog Using WordPress

Aim

To create a web blog using WordPress and demonstrate HTML elements such as Anchor Tag, Title Tag, headings, paragraphs, and images.

Algorithm

1. Start WordPress.
2. Create a new website or blog.
3. Select a suitable theme.
4. Create a new blog post.
5. Add a suitable title using the Title Tag.
6. Add headings and paragraph content.
7. Insert an image into the blog.
8. Create a hyperlink using the Anchor Tag.
9. Preview and publish the blog.
10. Verify that all elements are displayed correctly.

Program / HTML Example

```
<!DOCTYPE html>

<html>

<head>

    <title>My Web Blog</title>

</head>


<body>


    <h1>Welcome to My Web Blog</h1>


    <h2>About My Blog</h2>


    <p>

        This blog contains information about technology and programming.

    </p>


    <a href="https://www.google.com">Visit Google</a>
```

```
    <br><br>

    <img src="blog.jpg" width="300" height="200">

</body>
</html>
```

Output

```
------------------------------------------------

              Welcome to My Web Blog

About My Blog

This blog contains information about technology
and programming.

Visit Google

        [ Blog Image ]
------------------------------------------------
```

Result

Thus, a web blog was successfully created using WordPress, demonstrating the **Title Tag, Heading Tags, Paragraph Tag, Anchor Tag, and Image Tag**.


SWI-Prolog -- C:/Users/welcome/Desktop/experiment31.pl
File Settings Tools Debug GUI Help
73 ?- [experiment31].
true.
74 ?- display.
Blog: my_blog
Title: My Web Blog
Heading: Welcome to My Web Blog
Paragraph: This blog contains information about technology and programming.
Link: Visit Google
URL: https://www.google.com
Image: blog.jpg
true.